

Types of Data

- Numerical Data
- Categorical Data

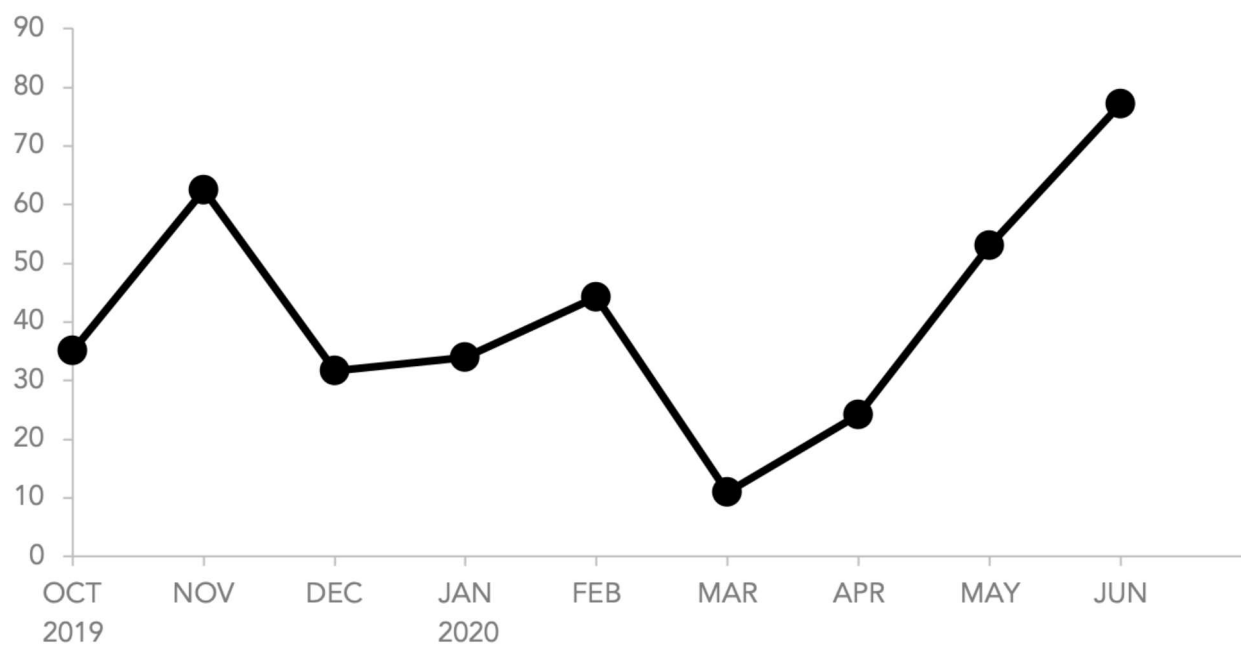
```
# import the library
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('default')
```

2D Line plot

Produce sales

IN THOUSANDS (USD)

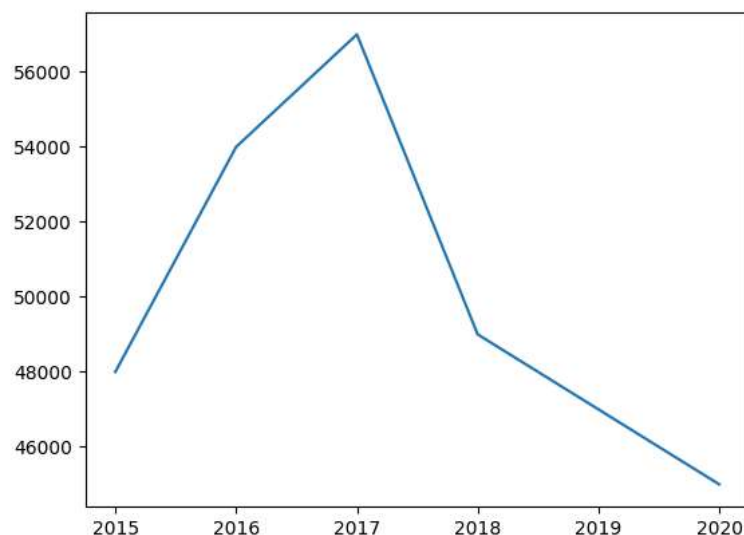


- Bivariate Analysis
- categorical -> numerical and numerical -> numerical
- Use case - Time series data

```
# plotting a simple function
price = [48000, 54000, 57000, 49000, 47000, 45000]
year = [2015, 2016, 2017, 2018, 2019, 2020]

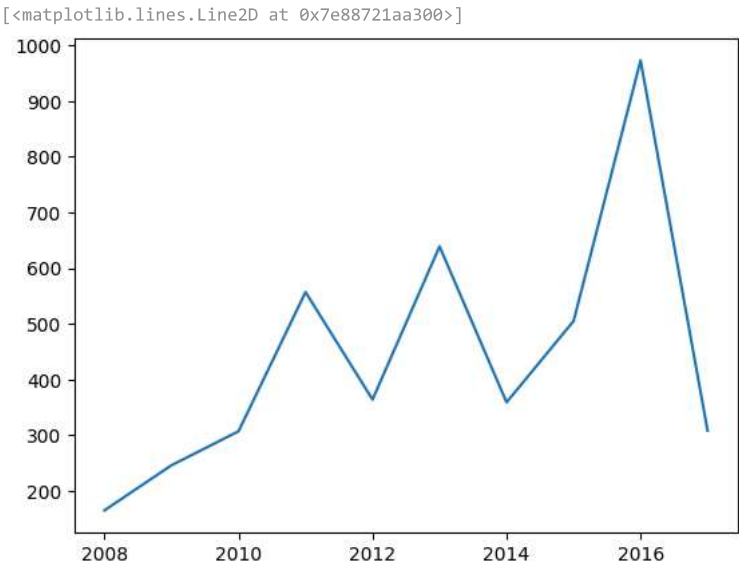
plt.plot(year, price)
```

```
[<matplotlib.lines.Line2D at 0x7e88723107a0>]
```

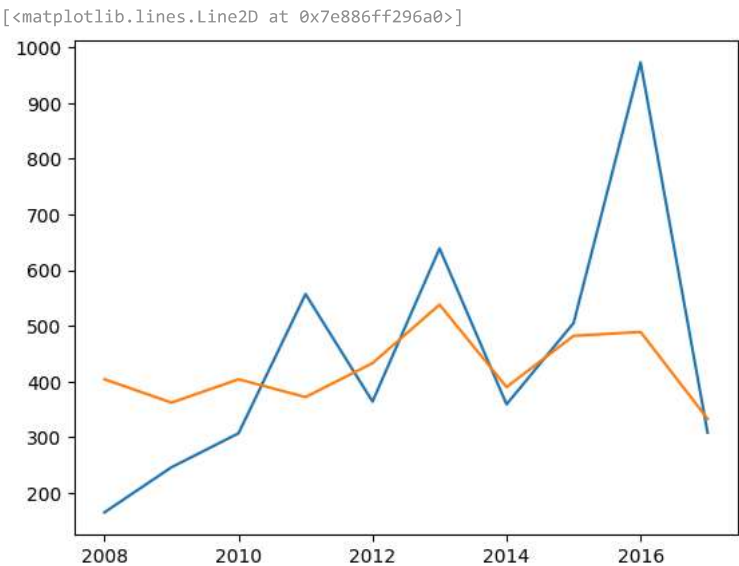


```
# from a pandas dataframe
batsman = pd.read_csv('/content/sharma-kohli.csv')
batsman
```

```
plt.plot(batsman['index'],batsman['V Kohli'])
```

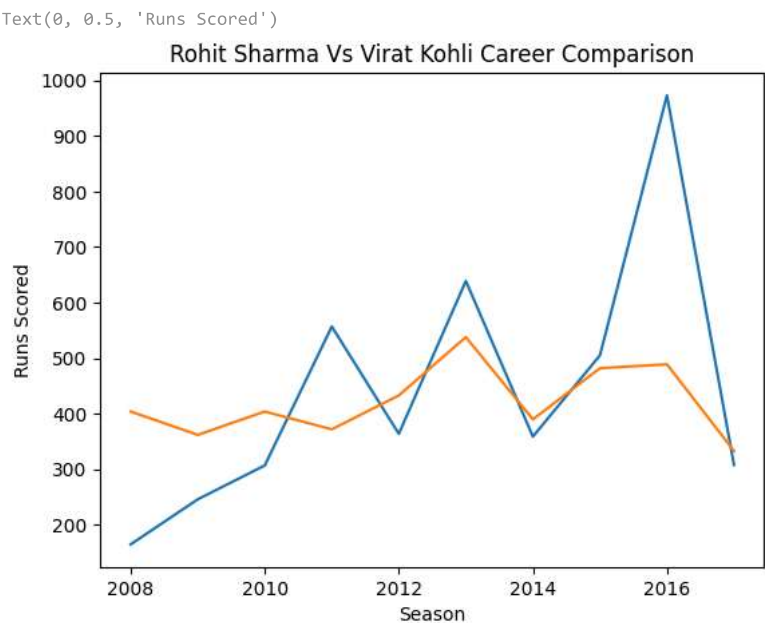


```
# plotting multiple plots
plt.plot(batsman['index'],batsman['V Kohli'])
plt.plot(batsman['index'],batsman['RG Sharma'])
```



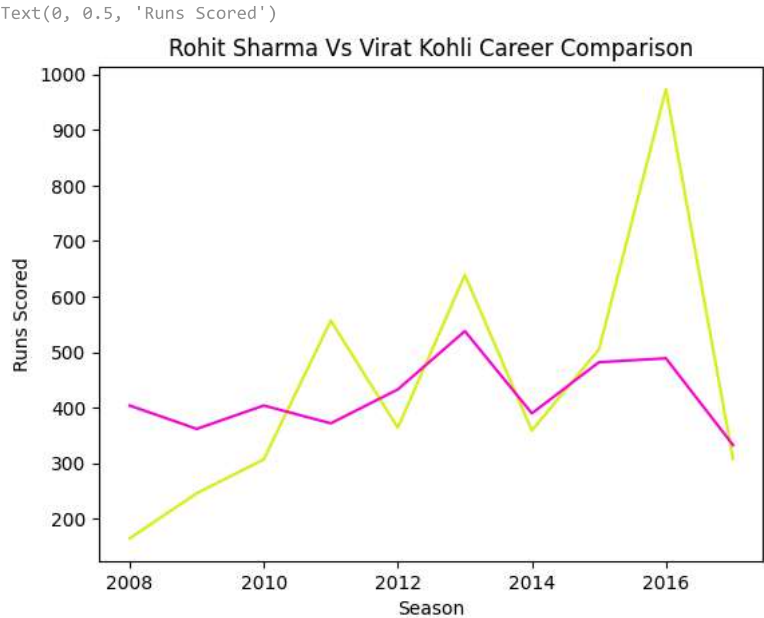
```
# labels title
plt.plot(batsman['index'],batsman['V Kohli'])
plt.plot(batsman['index'],batsman['RG Sharma'])

plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')
```



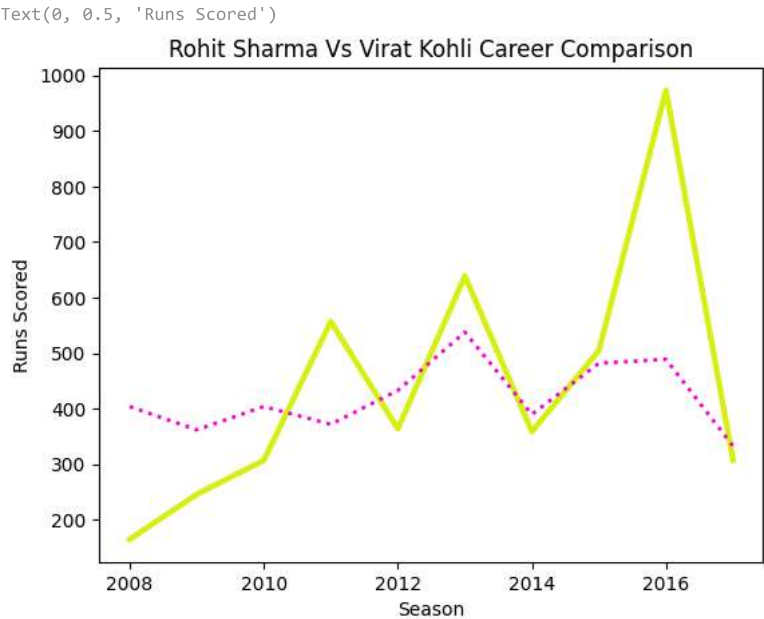
```
# colors(hex) and line(width and style) and marker(size)
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F')
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6')
```

```
plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')
```



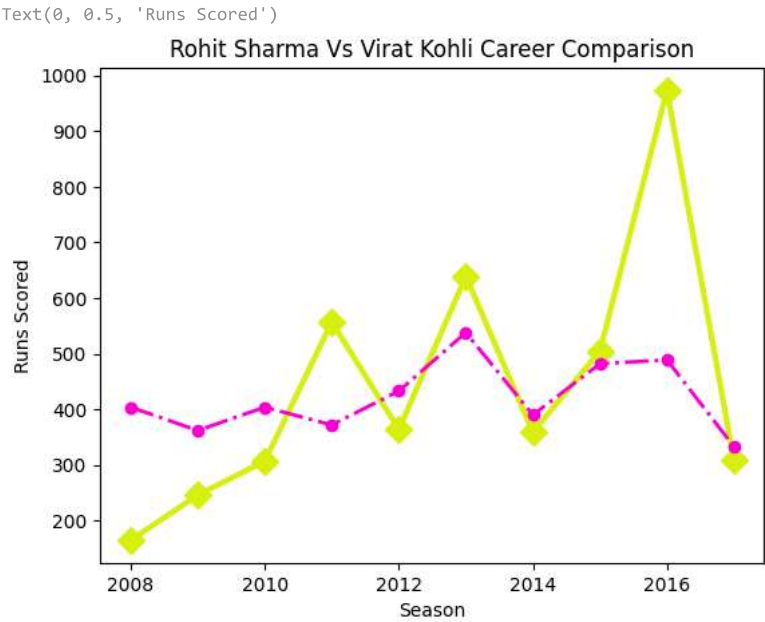
```
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F',linestyle='solid',linewidth=3)
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6',linestyle='dotted',linewidth=2)

plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')
```



```
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F',linestyle='solid',linewidth=3,marker='D',markersize=10)
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6',linestyle='dashdot',linewidth=2,marker='o')

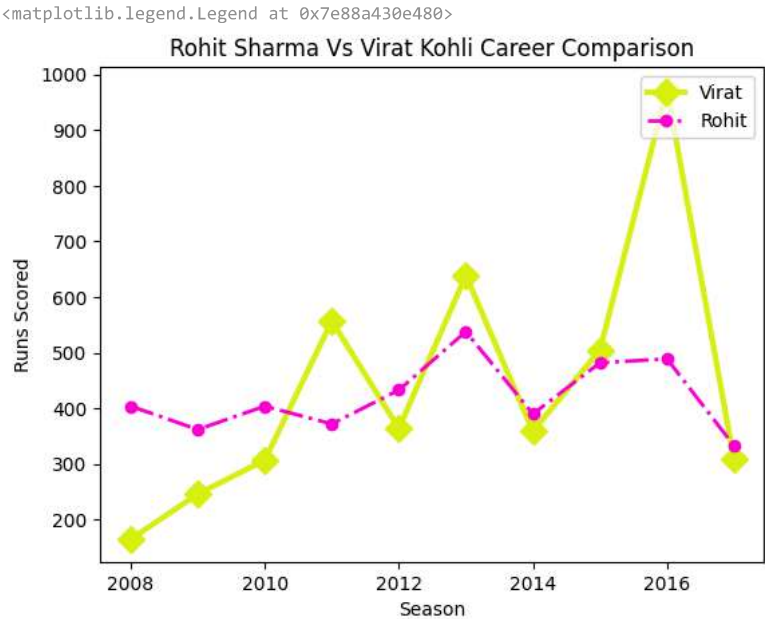
plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')
```



```
# legend -> location
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F',linestyle='solid',linewidth=3,marker='D',markersize=10,label='Virat')
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6',linestyle='dashdot',linewidth=2,marker='o',label='Rohit')

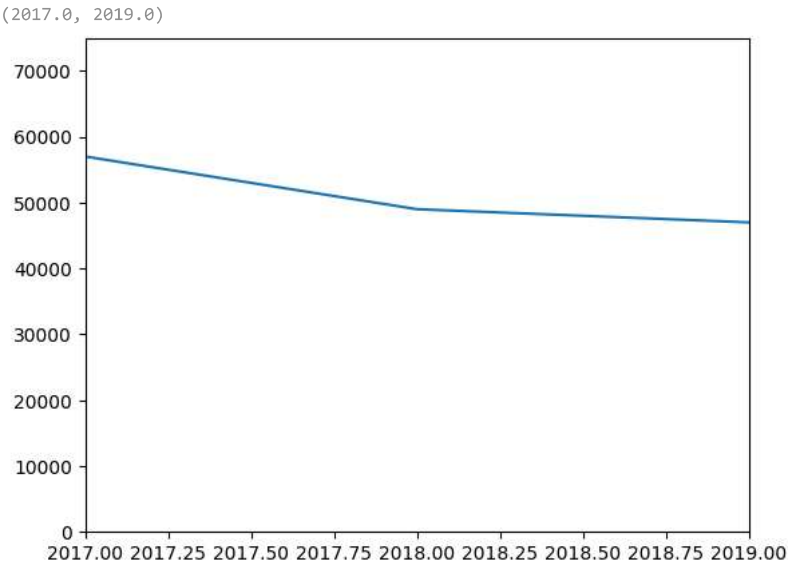
plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')

plt.legend(loc='upper right')
```



```
# limiting axes
price = [48000,54000,57000,49000,47000,45000,4500000]
year = [2015,2016,2017,2018,2019,2020,2021]

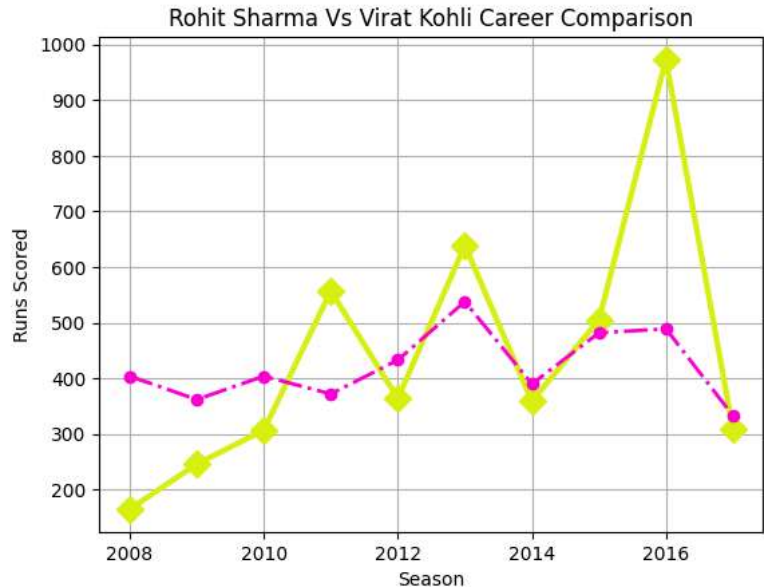
plt.plot(year,price)
plt.ylim(0,75000)
plt.xlim(2017,2019)
```



```
# grid
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F',linestyle='solid',linewidth=3,marker='D',markersize=10)
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6',linestyle='dashdot',linewidth=2,marker='o')

plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')

plt.grid()
```

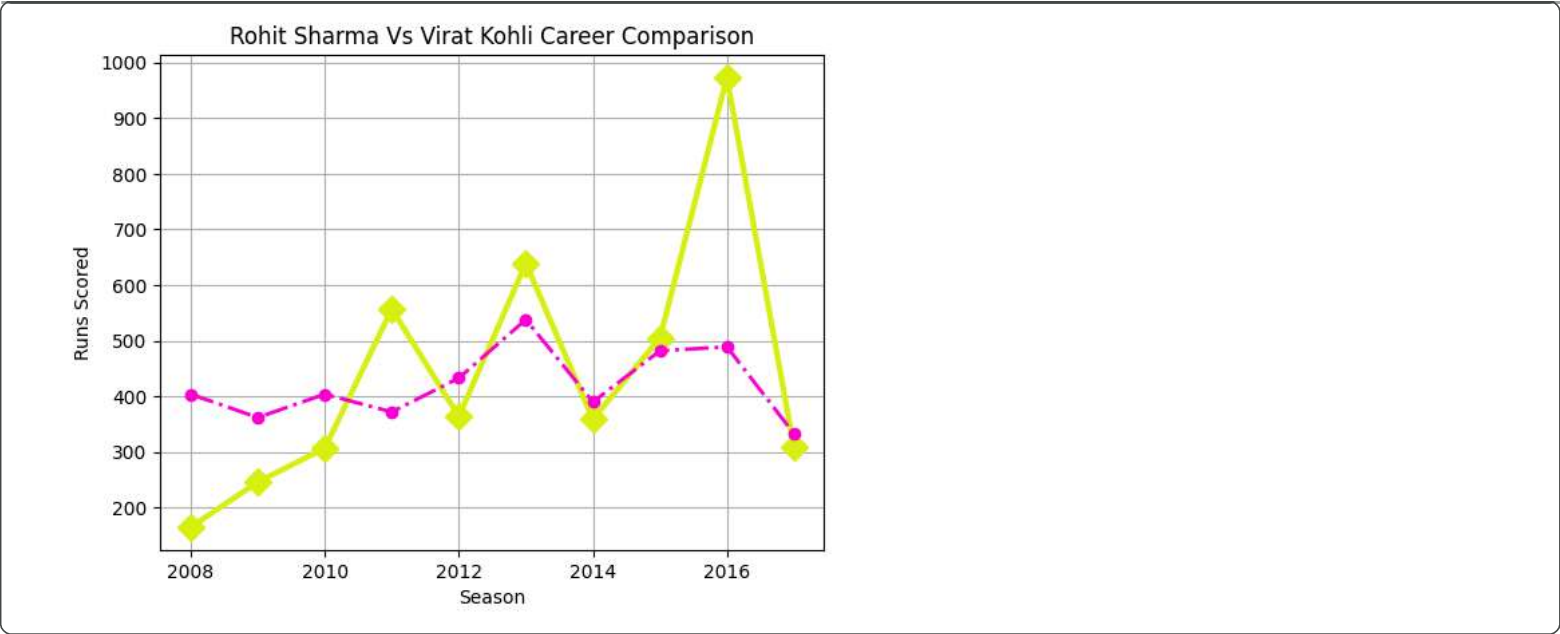


```
# show
plt.plot(batsman['index'],batsman['V Kohli'],color='#D9F10F',linestyle='solid',linewidth=3,marker='D',markersize=10)
plt.plot(batsman['index'],batsman['RG Sharma'],color='#FC00D6',linestyle='dashdot',linewidth=2,marker='o')

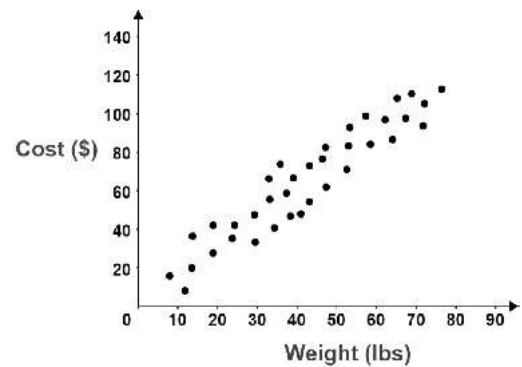
plt.title('Rohit Sharma Vs Virat Kohli Career Comparison')
plt.xlabel('Season')
plt.ylabel('Runs Scored')

plt.grid()

plt.show()
```



Scatter Plots



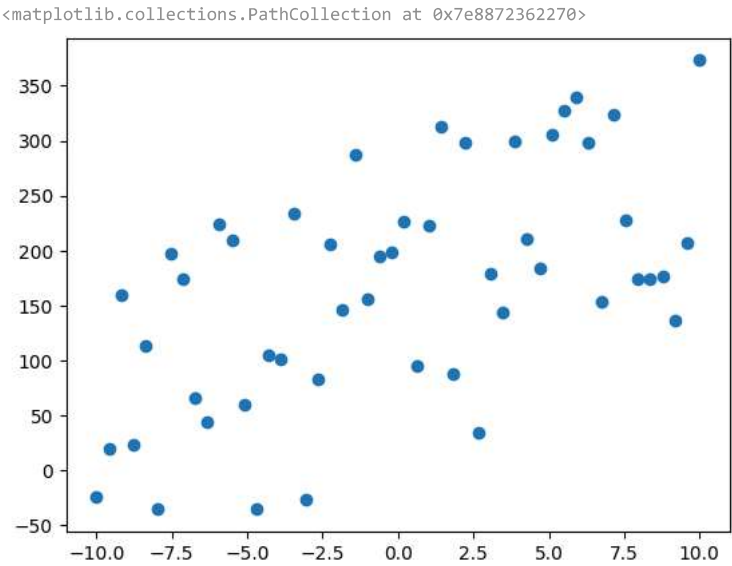
- Bivariate Analysis
- numerical vs numerical
- Use case - Finding correlation

```
# plt.scatter simple function
x = np.linspace(-10,10,50)

y = 10*x + 3 + np.random.randint(0,300,50)
y

array([-24.          , 20.08163265, 159.16326531, 23.24489796,
       113.32653061, -35.59183673, 197.48979592, 173.57142857,
        65.65306122,  43.73469388, 223.81632653, 208.89795918,
        59.97959184, -34.93877551, 105.14285714, 101.2244898 ,
       234.30612245, -26.6122449 ,  82.46938776, 205.55102041,
       145.63265306, 286.71428571, 155.79591837, 194.87755102,
       197.95918367, 226.04081633,  95.12244898, 222.20408163,
       312.28571429,  88.36734694, 298.44897959,  34.53061224,
       178.6122449 , 143.69387755, 299.7755102 , 210.85714286,
       183.93877551, 305.02040816, 327.10204082, 339.18367347,
       298.26530612, 153.34693878, 323.42857143, 227.51020408,
       173.59183673, 174.67346939, 176.75510204, 136.83673469,
       206.91836735, 373.          ])
```

```
plt.scatter(x,y)
```

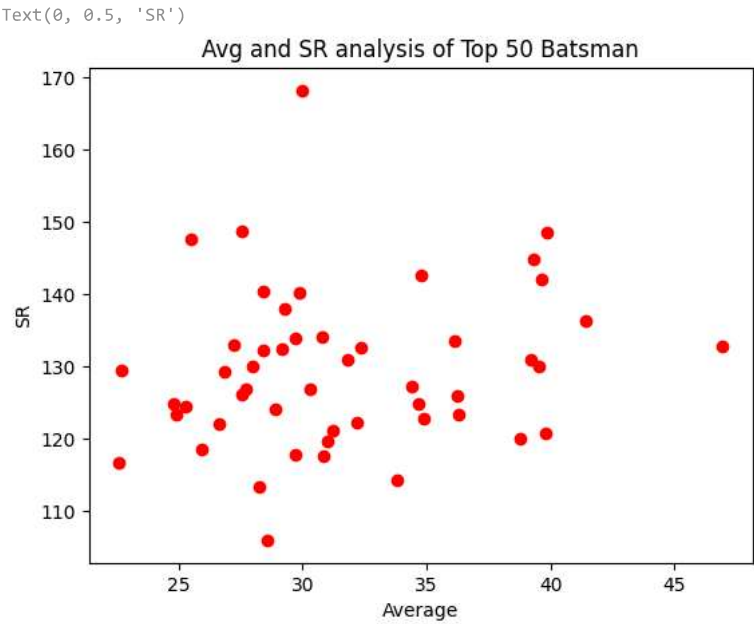


```
# plt.scatter on pandas data
df = pd.read_csv('/content/batter.csv')
df = df.head(50)
df.head()
```

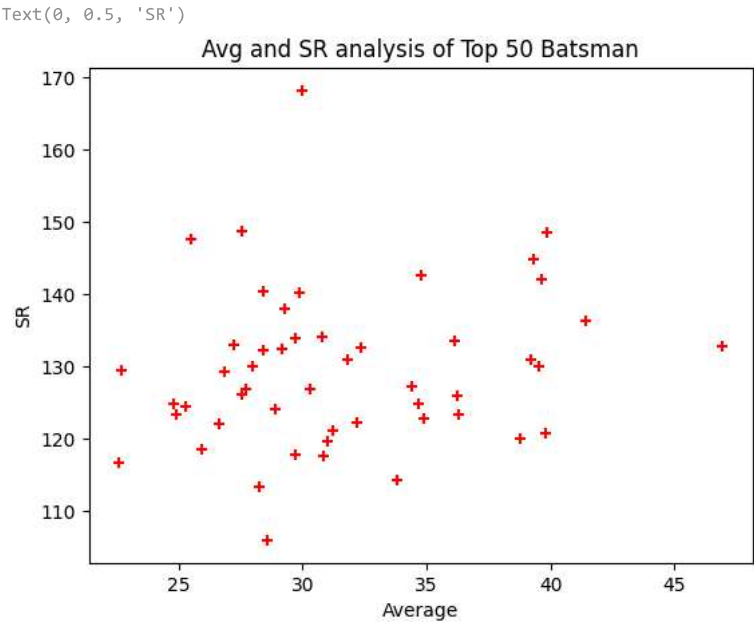
	batter	runs	avg	strike_rate
0	V Kohli	6634	36.251366	125.977972
1	S Dhawan	6244	34.882682	122.840842
2	DA Warner	5883	41.429577	136.401577
3	RG Sharma	5881	30.314433	126.964594
4	SK Raina	5536	32.374269	132.535312

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
plt.scatter(df['avg'],df['strike_rate'],color='red')
plt.title('Avg and SR analysis of Top 50 Batsman')
plt.xlabel('Average')
plt.ylabel('SR')
```

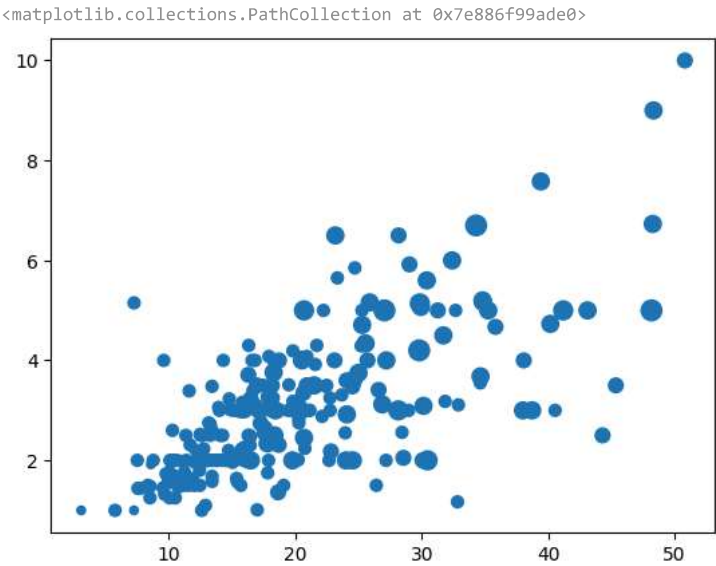


```
# marker
plt.scatter(df['avg'],df['strike_rate'],color='red',marker='+')
plt.title('Avg and SR analysis of Top 50 Batsman')
plt.xlabel('Average')
plt.ylabel('SR')
```

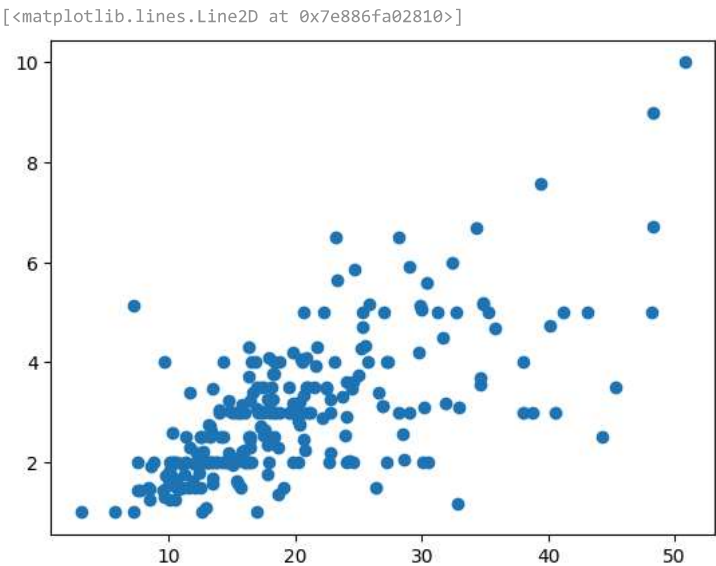


```
# size
tips = sns.load_dataset('tips')

# slower
plt.scatter(tips['total_bill'],tips['tip'],s=tips['size']*20)
```

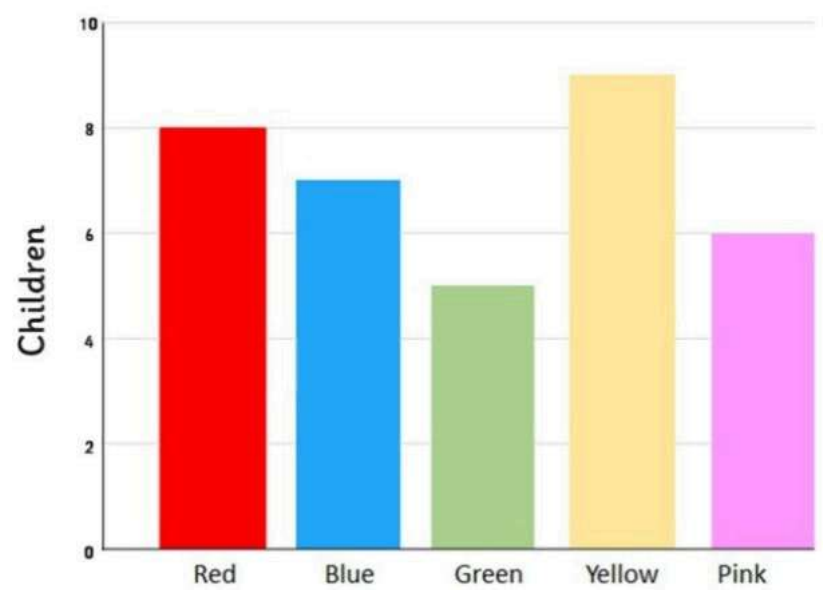


```
# scatterplot using plt.plot
# faster
plt.plot(tips['total_bill'],tips['tip'],'o')
```



```
# plt.plot vs plt.scatter
```

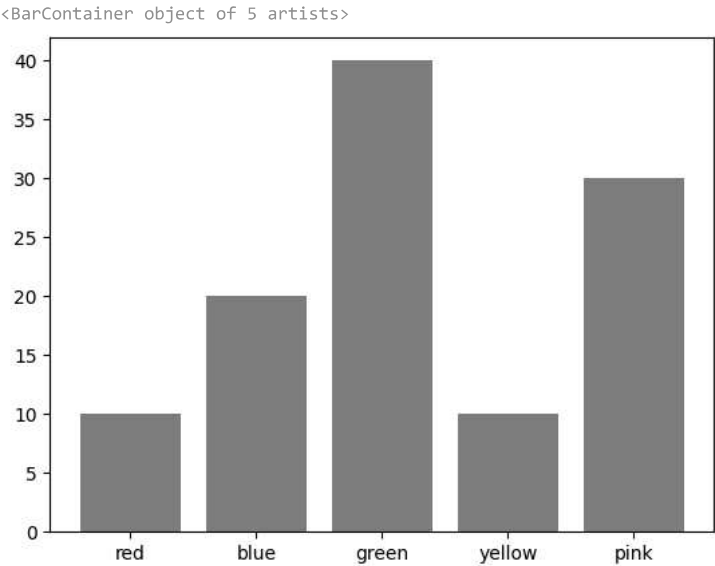

Favourite Colour



- Bivariate Analysis
- Numerical vs Categorical
- Use case - Aggregate analysis of groups

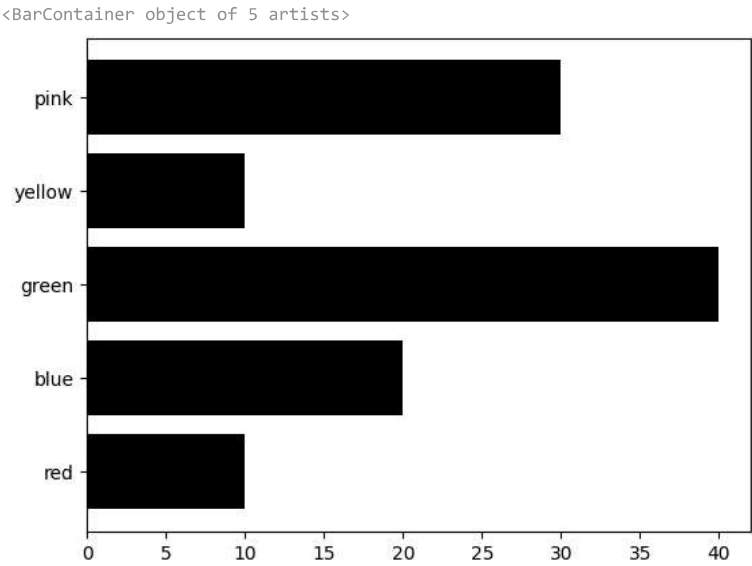
```
# simple bar chart
children = [10,20,40,10,30]
colors = ['red','blue','green','yellow','pink']

plt.bar(colors,children,color='grey')
```



```
# bar chart using data
```

```
# horizontal bar chart
plt.barh(colors,children,color='black')
```



```
# color and label
df = pd.read_csv('/content/batsman_season_record.csv')
df
```

	batsman	2015	2016	2017
0	AB de Villiers	513	687	216
1	DA Warner	562	848	641
2	MS Dhoni	372	284	290
3	RG Sharma	482	489	333
4	V Kohli	505	973	308

Next steps:

Generate code with df

New interactive sheet

```
# Filter Dhoni row
dhoni = df[df['batsman'] == 'MS Dhoni']

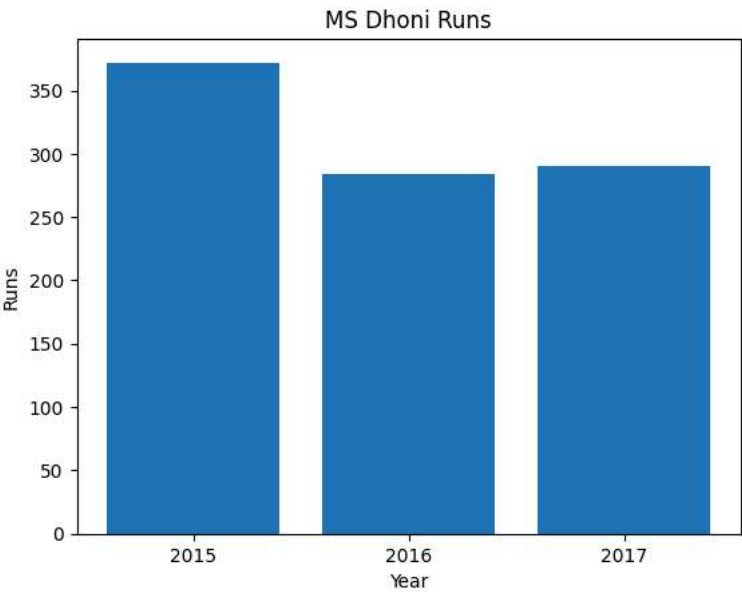
# Years
years = ['2015', '2016', '2017']

# Runs
runs = dhoni[years].values[0]

# Bar chart
plt.bar(years, runs)

plt.title('MS Dhoni Runs')
plt.xlabel('Year')
plt.ylabel('Runs')

plt.show()
```



```
np.arange(df.shape[0])
```

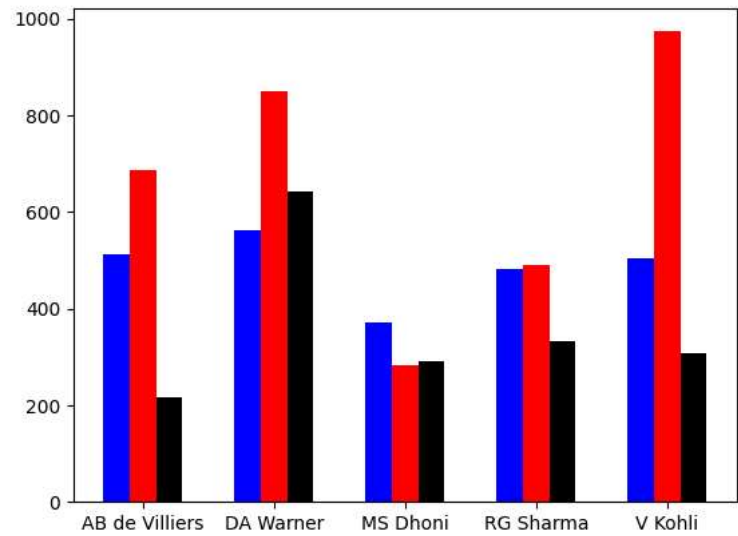
```
array([0, 1, 2, 3, 4])
```

```
# Multiple Bar charts
plt.bar(np.arange(df.shape[0]) - 0.2,df['2015'],width=0.2,color='blue')
```

```
plt.bar(np.arange(df.shape[0]),df['2016'],width=0.2,color='red')
plt.bar(np.arange(df.shape[0]) + 0.2,df['2017'],width=0.2,color='black')

plt.xticks(np.arange(df.shape[0]), df['batsman'])

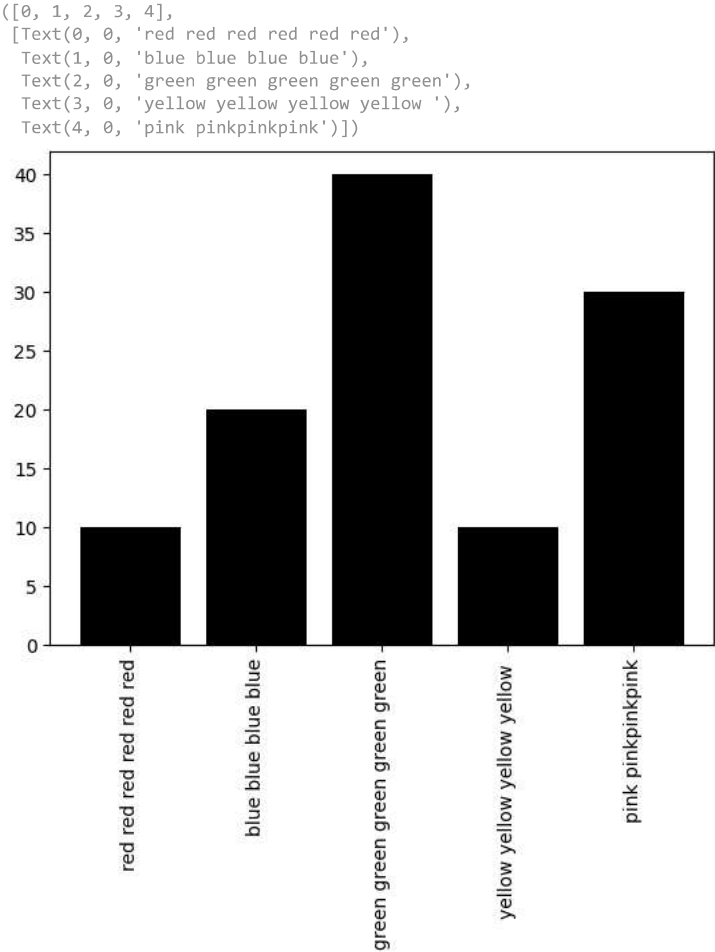
plt.show()
```



xticks

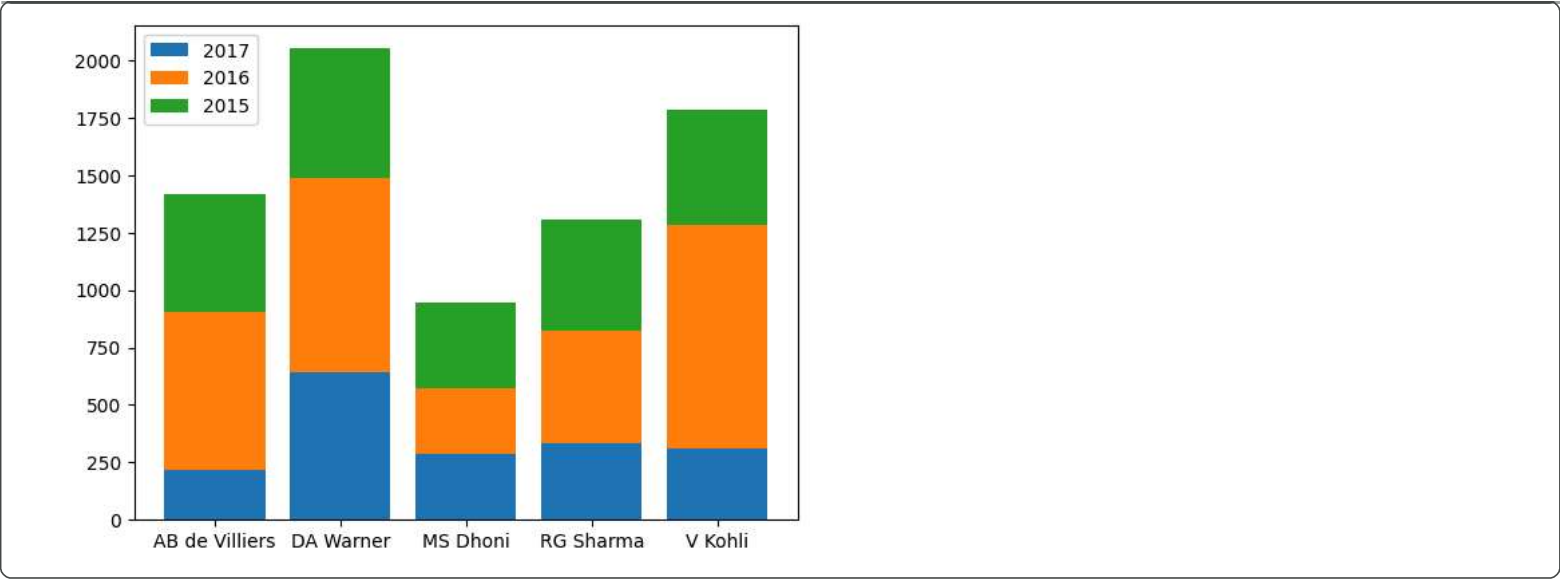
```
# a problem
children = [10,20,40,10,30]
colors = ['red red red red red red','blue blue blue blue','green green green green green','yellow yellow yellow yellow ','pink pinkpinkpink']

plt.bar(colors,children,color='black')
plt.xticks(rotation='vertical')
```



```
# Stacked Bar chart
plt.bar(df['batsman'],df['2017'],label='2017')
plt.bar(df['batsman'],df['2016'],bottom=df['2017'],label='2016')
plt.bar(df['batsman'],df['2015'],bottom=(df['2016'] + df['2017']),label='2015')

plt.legend()
plt.show()
```



▼ Histogram



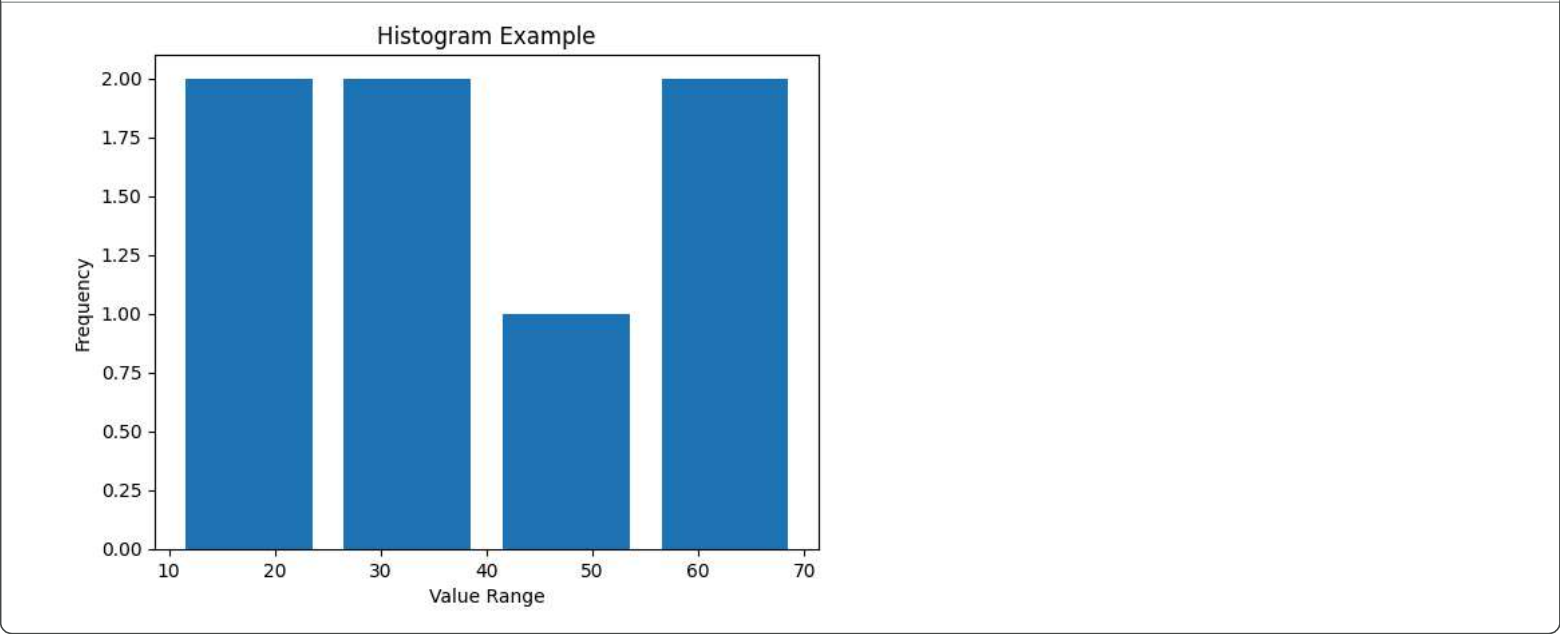
- Univariate Analysis
- Numerical col
- Use case - Frequency Count

```
# simple data
data = [32,45,56,10,15,27,61]

# histogram
plt.hist(data, bins=[10,25,40,55,70],rwidth=0.8)

plt.xlabel("Value Range")
plt.ylabel("Frequency")
plt.title("Histogram Example")

plt.show()
```



```
# on some data
df = pd.read_csv('/content/vk.csv')
df
```

	match_id	batsman_runs
0	12	62
1	17	28
2	20	64
3	27	0
4	30	10
...	...	...
136	624	75
137	626	113
138	632	54
139	633	0
140	636	54

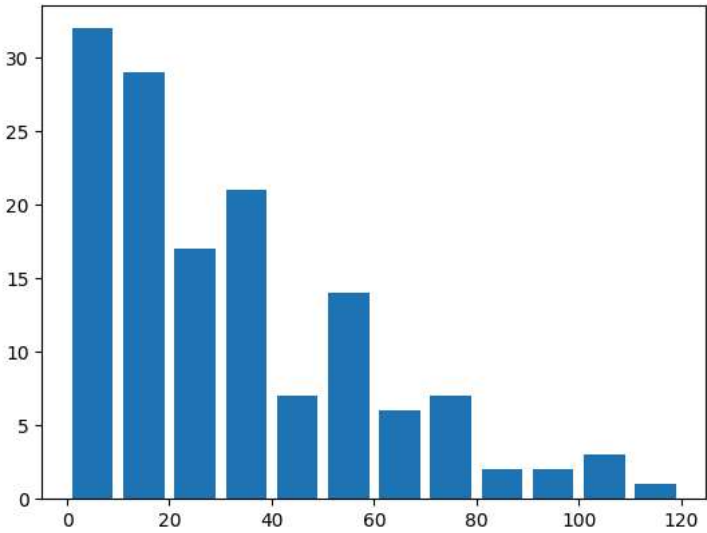
141 rows × 2 columns

Next steps:

Generate code with df

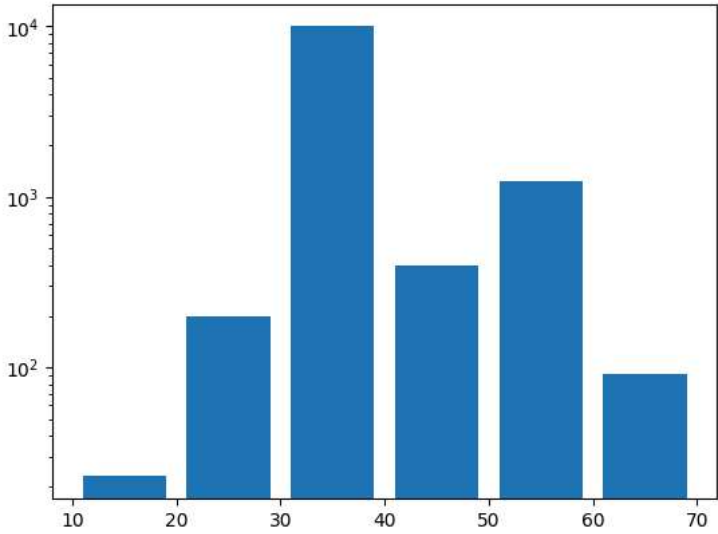
New interactive sheet

```
plt.hist(df['batsman_runs'],bins=[0,10,20,30,40,50,60,70,80,90,100,110,120],rwidth=0.8)
plt.show()
```

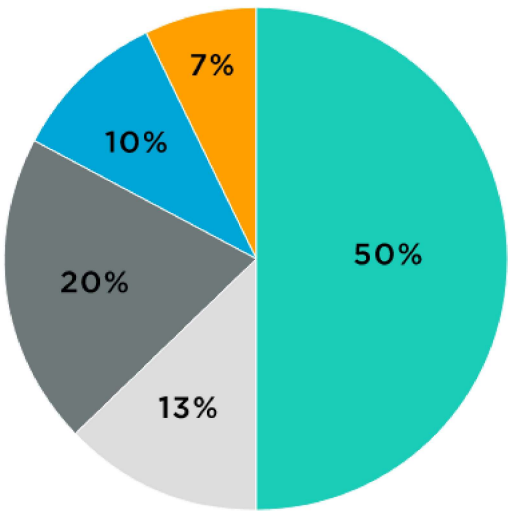


handling bins

```
# logarithmic scale
arr = np.load('/content/big-array.npy')
plt.hist(arr,bins=[10,20,30,40,50,60,70],log=True,rwidth=0.8)
plt.show()
```



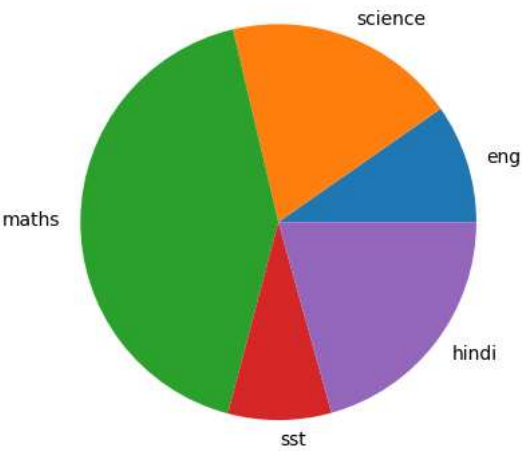
⌵ Pie Chart



- Univariate/Bivariate Analysis
- Categorical vs numerical
- Use case - To find contibution on a standard scale

```
# simple data
data = [23,45,100,20,49]
subjects = ['eng','science','maths','sst','hindi']
plt.pie(data,labels=subjects)

plt.show()
```

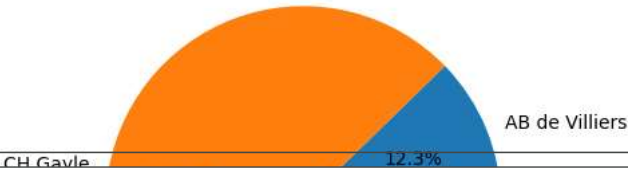


```
# dataset
df = pd.read_csv('/content/gayle-175.csv')
df
```

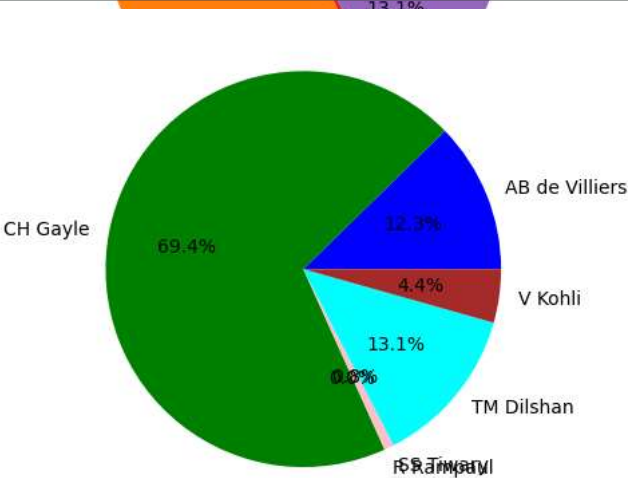
	batsman	batsman_runs
0	AB de Villiers	31
1	CH Gayle	175
2	R Rampaul	0
3	SS Tiwary	2
4	TM Dilshan	33
5	V Kohli	11

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
plt.pie(df['batsman_runs'],labels=df['batsman'],autopct='%0.1f%%')
plt.show()
```



```
# percentage and colors
plt.pie(df['batsman_runs'],labels=df['batsman'],autopct='%0.1f%%',colors=['blue','green','yellow','pink','cyan','brown'])
plt.show()
```



```
# explode shadow
plt.pie(df['batsman_runs'],labels=df['batsman'],autopct='%0.1f%%',explode=[0.3,0,0,0,0,0.1],shadow=True)
plt.show()
```